

BTEC Level 3 Applied Science • Unit 1 Chemistry • Worksheet

Covalent Bonding

Original JDScience practice. These questions were written for this resource and are not official exam-board items.

A Retrieval

1. Define a covalent bond.
2. What is a lone pair?
3. Give one simple-molecular and one giant-covalent example.

B Knowledge

4. Explain how a chlorine molecule is held together.
5. Describe the bonding and structure of diamond.
6. Why does graphite conduct electricity when diamond does not?

C Application

7. State the number of shared pairs in F₂, O₂, N₂ and CH₄.
8. Explain why water has a much lower boiling point than silicon dioxide.
9. Why does boiling water not break O–H covalent bonds?

D Exam-style practice

10. Describe how a covalent bond forms in a hydrogen molecule. [3]
11. Explain why iodine melts far below diamond. [4]
12. Explain why carbon dioxide does not conduct electricity. [2]

E Challenge

13. Ammonia can accept a proton to form NH₄⁺. Explain this using the idea of a lone pair and a dative covalent bond.